

Lab-13 Ex1 - A Recursive GCD Solver

In Lab-8 Ex3, you have seen a Greatest Common Divisor (GCD) solver using the Euclidean algorithm https://en.wikipedia.org/wiki/Euclidean_algorithm#Implementations. Besides the iterative approach, we can also use recursion to find GCD of two integers.

The pseudocode is very simple:

```
function gcd(x,y)
    if y is 0, return x
    otherwise, return gcd(y, remainder of x/y)
```

Are you able to identify the **termination condition** and the **function call** of itself, which are required components of a recursive function?

Now, write a simple C program to ask user for x and y . You can assume them to be positive integers. Then output the GCD of these two numbers.

User input is ***bold and italic*** in the following sample runs.

Sample Run #1

x? ***27***

y? ***9***

GCD of 27 and 9 is 9

Sample Run #2

x? ***111***

y? ***259***

GCD of 111 and 259 is 37

Note: you can read more about the recursive version of Euclidean algorithm here: [https://en.wikipedia.org/wiki/Recursion_\(computer_science\)#Greatest_common_divisor](https://en.wikipedia.org/wiki/Recursion_(computer_science)#Greatest_common_divisor)